DL-BASED PUBLIC KEY ENCRYPTION SCHEMES

Diffie-Hellman Key Exchange

- Let $G = \langle g \rangle$ be a cyclic group of order m.

Alice		Bob
$x \xleftarrow{\$} \mathbb{Z}_m; X \leftarrow g^x$	$\xrightarrow{X}$ $\xleftarrow{Y}$	$y \xleftarrow{\$} \mathbb{Z}_m; Y \leftarrow g^y$

- Then

$$Y^x = (g^y)^x = g^{xy} = (g^x)^y = X^y$$

  - Alice can compute $K = Y^x$
  - Bob can compute $K = X^y$
  - But adversary wanting to compute K is faced with

$$g^x, g^y \longrightarrow g^{xy}$$

    which is exactly the CDH problem and is believed to be hard

ElGamal Scheme Idea

- We can turn DH key exchange into a public key encryption scheme via
  - Let Alice have public key g^x and secret key x
  - If Bob wants to encrypt M for Alice, he
    - Picks y and sends g^y to Alice
    - Encrypts M under $g^{xy} = (g^x)^y$ and sends ciphertext to Alice.
  - But Alice can re-compute $g^{xy} = (g^y)^x$ because she knows
    - g^y
    - x is her secret key

    Thus she can decrypt and adversary is still faced with CDH.

The ElGamal Scheme

- Let G be a cyclic group of order m and let g be a generator of G The **ElGamal** encryption scheme $\mathrm{EG} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ associated to G, g is as follows:

Algorithm $\mathcal{K}$

$x \xleftarrow{\$} \mathbb{Z}_m$

$X \leftarrow g^x$

return (X, x)

Algorithm $\mathcal{E}_X(M)$

$y \xleftarrow{\$} \mathbb{Z}_m; Y \leftarrow g^y$

$K \leftarrow X^y$

$W \leftarrow K \cdot M$

return (Y, W)

Algorithm $\mathcal{D}_x(Y, W)$

$K = Y^x$

$M \leftarrow W \cdot K^{-1}$

return M

Security of ElGamal

secret key $= x \in \mathbb{Z}_m$, where $m = |G|$

public key $= X = g^x \in G = \langle g \rangle$

Algorithm $\mathcal{E}_X(M)$	**Algorithm** $\mathcal{D}_x(Y, W)$
$y \xleftarrow{\$} \mathbb{Z}_m; Y \leftarrow g^y$	$K = Y^x; M \leftarrow W \cdot K^{-1}$
$K \leftarrow X^y; W \leftarrow K \cdot M$	return M
return (Y, W)	

- To find x given X, adversary must solve **DL** problem
- To find M given $X, (Y, W)$, adversary must compute $K = g^{xy}$, meaning solve **CDH** problem
- But what prevents leakage of partial information about M? Is the scheme IND-CPA secure?

Insecurity of ElGamal in $\mathbb{Z}_p^*$

- ElGamal in $\mathbb{Z}_p^*$ is **not** IND-CPA secure.
- The reason is the same as why the DDH is easy in $\mathbb{Z}_p^*$: squares vs. non-squares.

Insecurity of ElGamal in $\mathbb{Z}_p^*$

- Let p be a prime and g a generator of $\mathbb{Z}_p^*$. The EG PKE scheme $\mathcal{AE}_{\text{EG}} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ is defined by

Algorithm $\mathcal{K}$	**Algorithm** $\mathcal{E}_X(M)$	**Algorithm** $\mathcal{D}_x(Y, W)$
$x \xleftarrow{\$} \mathbb{Z}_{p-1}$	$y \xleftarrow{\$} \mathbb{Z}_{p-1}; Y \leftarrow g^y$	$K = Y^x$
$X \leftarrow g^x$	$K \leftarrow X^y$	$M \leftarrow W \cdot K^{-1}$
return (X, x)	$W \leftarrow K \cdot M$	return M
	return (Y, W)	

- **The weakness:** Suppose $(Y, W) \xleftarrow{\$} \mathcal{E}_X(M)$. Then we claim that given
  - the public key X
  - the ciphertext (Y, W)

  an adversary can easily compute $J_p(M)$.

  This represents a loss of partial information.

Insecurity of ElGamal in $\mathbb{Z}_p^*$

- Let p be a prime and g a generator of $\mathbb{Z}_p^*$. The EG PKE scheme $\mathcal{AE}_{\text{EG}} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ is defined by

Algorithm $\mathcal{K}$

$x \xleftarrow{\$} \mathbb{Z}_{p-1}$
$X \leftarrow g^x$
return (X, x)

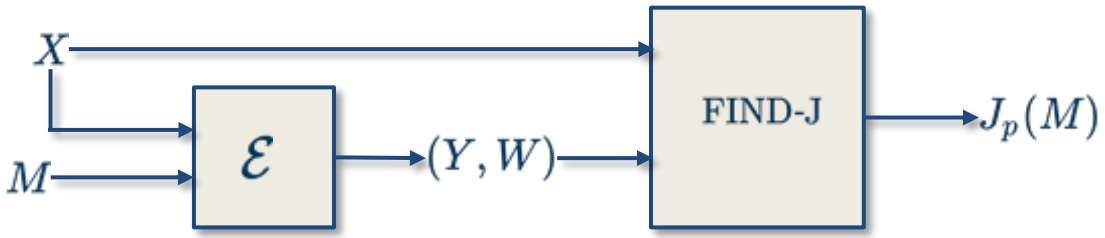
Algorithm $\mathcal{E}_X(M)$

$y \xleftarrow{\$} \mathbb{Z}_{p-1}; Y \leftarrow g^y$
$K \leftarrow X^y$
$W \leftarrow K \cdot M$
return (Y, W)

Algorithm $\mathcal{D}_x(Y, W)$

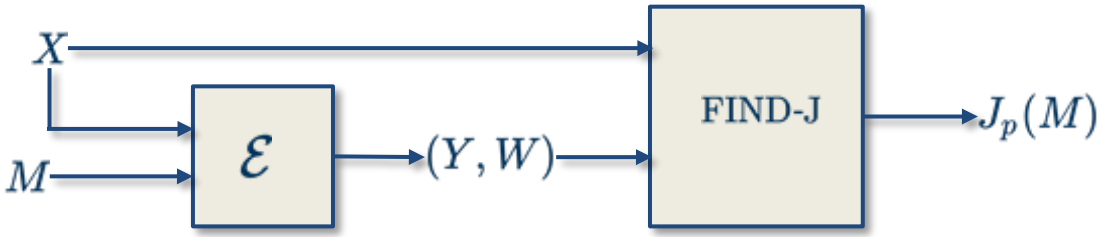
$K = Y^x$
$M \leftarrow W \cdot K^{-1}$
return M

- **The weakness:** There is an algorithm FIND-J



Attack Idea

- Given public key X
  - Let M_0, M_1 be such that $J_p(M_0) = -1$ and $J_p(M_1) = 1$
  - Receive encryption (Y, W) of M_b



  - If $\text{FIND-J}(X, Y, W) = 1$ then return 1 else return 0

IND-CPA Attack

Algorithm FIND-J(X, Y, W)

if $J_p(X) = 1$ or $J_p(Y) = 1$ then $s \leftarrow 1$ else $s \leftarrow -1$

return $J_p(W) \cdot s$

Adversary $\mathcal{A}(X)$

$M_1 \leftarrow 1; M_0 \leftarrow g$

$(Y, W) \xleftarrow{\$} LR(M_0, M_1)$

if Find-J$(X, Y, W) = 1$ then return 1 else return 0

Security of ElGamal

- We have seen that EG is not IND-CPA over groups $G = \mathbb{Z}_p^*$ for prime p. However it is IND-CPA secure over any group G where the DDH problem is hard.

- We can in particular securely implement ElGamal over
  - Appropriate prime-order subgroups of $\mathbb{Z}_p^*$ for a prime p .
  - Elliptic curve groups of prime order

Security of ElGamal

- **Claim:** If DDH is hard in G then $\mathcal{AE}_{\mathrm{EG}}$ is IND-CPA secure.
- **Note:** DDH is NOT hard in $\mathbb{Z}_p^*$ (p is a prime).

DDH is Hard $\Rightarrow$ ElGamal is IND-CPA

- **Proof:**

ElGamal is Not IND-CCA

- **Proof.**

Cramer-Shoup Encryption

- Let G be a cyclic group of order q, and g_1, g_2 are random, distinct generators of G:

Algorithm $\mathcal{K}$

$x_1, x_2, y_1, y_2, z \xleftarrow{\$} \mathbb{Z}_q$

$c \leftarrow g_1^{x_1} g_2^{x_2}; d \leftarrow g_1^{y_1} g_2^{y_2}; h \leftarrow g_1^{z}$

return $(G, q, g_1, g_2, c, d, h, x_1, x_2, y_1, y_2, z)$

Algorithm $\mathcal{E}_{(G,q,g_1,g_2,c,d,h)}(M)$

$k \xleftarrow{\$} \mathbb{Z}_q$

$u_1 \leftarrow g_1^k; u_2 \leftarrow g_2^k$

$e \leftarrow h^k \cdot M$

$\alpha \leftarrow H(u_1, u_2, e)$

$v \leftarrow c^k \cdot d^{k\alpha}$

return (u_1, u_2, e, v)

Algorithm $\mathcal{D}_{x_1,x_2,y_1,y_2,z}(\hat{u_1}, \hat{u_2}, \hat{e}, \hat{v})$

$\hat{\alpha} \leftarrow H(\hat{u_1}, \hat{u_2}, \hat{e})$;

If $\hat{u_1}^{x_1} \hat{u_1}^{x_2} (\hat{u_1}^{y_1} \hat{u_2}^{y_2})^{\hat{\alpha}} = \hat{v}$ then

  return $(\hat{e}/(\hat{u_1}^{z}))$

else

  return $\perp$

- **Fact:** If DDH is hard and H is a CR hash function, then Cramer Shoup is IND-CCA.